

A Brief Discussion in Gaussian Integer

William

May 2025

1 Introduction

The Gaussian integer was discovered by *Carolus Fridericus Gauss* (certainly), and defined as

$$\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\},$$

such numbers played an essential role in Gauss's work, who established contemporary Algebraic number theory that bridged the gap in the study of binary quadratic forms and studied quadratic residues. Gauss's profound comprehension of the Ring $\mathbb{Z}[i]$ has allowed him to give his first proof of the famous quadratic reciprocity law. However, since the quadratic residue are relatively prevail, we can hardly provide any insightful arguments. Thus, we would discuss about the primary attributes of ring $\mathbb{Z}[i]$ that Gauss discovered in this article and indicate its relation to the quadratic forms.

2 Basic Property

In order to study $\mathbb{Z}[i]$, we have to define its Euclidean function and show that it is an Euclidean ring. The fractional field of the $\mathbb{Z}[i]$ is obviously $\mathbb{Q}[i]$. The Euclid function of a ring should depict the "magnitude" of the elements, and we recall the magnitude of the complex number. The conjugate of the complex number can be employed as well: $\overline{a + bi} = a - bi$. We define the Norm that measures the "distance".

Definition 2.1. For $a, b \in \mathbb{Q}$, $N(a + bi) := a^2 + b^2$

The norm is a complete multiplicative function. We immediately derive some properties of $N(\alpha)$ in $\mathbb{Z}[i]$:

Lemma 2.2. For $\alpha, \beta \in \mathbb{Z}[i]$, $N(\alpha\beta) = N(\alpha)N(\beta)$, and thus $\alpha \mid \beta \iff N(\alpha) \mid N(\beta)$.

Lemma 2.3. The $\alpha \in \mathbb{Z}[i]$ is the identity element $\iff N(\alpha) = 1$; thus the only identity elements in $\mathbb{Z}[i]$ is $1, i, -1, -i$.

Readers can prove these lemmas for themselves. The ring $\mathbb{Z}[i]$ is generated by $\{1, i, -1, -i\}$, so the associated elements ($\sim$) is clear. Because $N(\alpha)$ has limited number of factors, α must have limited factors as well. Therefore, the $\alpha \in \mathbb{Z}[i]$ can be factorized, and $\mathbb{Z}[i]$ is a factorial domain.

Now, we are about to determine the structure of principal ideal for Gaussian integers in order to prepare for stating the modular arithmetic in $\mathbb{Z}[i]$, which is the set of multiples of a random element Gaussian integer.

Theorem 2.4. If $\mu = a + bi \in \mathbb{Z}[i] \setminus \{0\}$ and $C, C' \in \mathbb{Z}$,

$$\theta \in (\mu) \iff \theta = \frac{CN(\mu)}{(a, b)} + C'\theta_0$$

for

$$\theta_0 = ax_0 - by_0 + (a, b)i, \quad ay_0 + bx_0 = (a, b)$$

Proof. If $\mu \mid \theta$, there is an $\delta = x + yi$ such that $\theta = \mu\delta$. We expand these numbers: $\theta = (ax - by) + (ay + bx)i$, and we give

$$C' = \frac{ay + bx}{(a, b)}$$

For the same θ_0 as in the theorem, the subtraction

$$\theta - C'\theta_0 = ax - by + (ay + bx)i - \frac{(ax_0 - by_0)(ay + bx)}{(a, b)} - (ay + bx)i = \frac{(xy_0 - yx_0)N(\mu)}{(a, b)},$$

since

$$ax - by - \frac{(ax_0 - by_0)(ay + bx)}{ay_0 + bx_0} = \frac{(xy_0 - yx_0)(a^2 + b^2)}{ay_0 + bx_0}.$$

We set $C = xy_0 - yx_0$ so that the θ is the same form as in the theorem. Using the construction, we can simply prove $\theta \in \langle \mu \rangle$. $\square$

3 $\mathbb{Z}[i]$'s modular system

First, we prove the $\mathbb{Z}[i]$ is an Euclidean domain.

Theorem 3.1. $\forall \alpha, \beta \in \mathbb{Z}[i], \exists \gamma, \delta \in \mathbb{Z}[i]$ such that

$$\beta = \delta\alpha + \gamma, 0 \leq N(\gamma) \leq N(\alpha)$$

, and $\gamma = 0 \iff \alpha \mid \beta$.

Proof. We rewrite the equation in theorem:

$$\frac{\beta}{\alpha} = \delta + \frac{\gamma}{\alpha}, 0 \leq N\left(\frac{\gamma}{\alpha}\right) < 1.$$

Since α, β is randomly selected, it is sufficient to show that $\forall \eta = a + bi \in \mathbb{Q}[i], \exists \delta \in \mathbb{Z}[i]$ such that $0 \leq N(\eta - \delta) < 1$. We can choose the nearest integer, c, d , to a, b which satisfies $\max(|a - c|, |b - d|) \leq \frac{1}{2}$, and set $\delta = c + di$. Thus, $N(\eta - \delta) = (a - c)^2 + (b - d)^2 \leq \frac{1}{2}$. $\square$

The Theorem has indicated the viability of Euclidean Algorithm in $\mathbb{Z}[i]$; the readers can verify by themselves. By the Euclidean Algorithm, we are able to prove

Theorem 3.2. $\mathbb{Z}[i]$ is a principal ideal domain.

Proof. Considering the set

$$A = \{N(\alpha) \mid \alpha \in S\}$$

, where S is an non-zero ideal. There is an α_0 such that $N(\alpha_0) = n_0 = \min A$. Then, any $\alpha \in S$ can be expressed as $\alpha = \beta\alpha_0 + \gamma$ for some $\beta, \gamma \in \mathbb{Z}[i]$, and $N(\gamma) < N(\alpha_0)$. According to the definition of ideal, the $\gamma \in S$, which leads to a contradiction. Therefore, $S = (\alpha_0)$. $\square$

That $\mathbb{Z}[i]$ is PID suffices to determine that $\mathbb{Z}[i]$ is Unique Factorized Domain (UFD). After we discover some great properties of the Gaussian integers as a domain, we may further propose some number-theoretic identities, which might be familiar to readers, such as the configuration of Gaussian primes.

Theorem 3.3. $\pi \in \mathbb{Z}[i]$ is a Gaussian primes if and only if it satisfies one of the following conditions:

$$\begin{cases} N(\pi) = 2 \\ N(\pi) = p \equiv 1 \pmod{4} \\ \pi \sim p \equiv 3 \pmod{4}, \end{cases}$$

where p is prime integer.

Proof. Sufficiency: $\forall N(\pi) \in \mathbb{P}$, π is non-factorized due to lemma 2.2. If $\pi \sim p \equiv 3 \pmod{4}$ and $\pi = \alpha\beta$ for some elements that are not identity, $N(\pi) = N(\alpha)N(\beta) = p^2$, so $N(\alpha) = N(\beta) = p$. For $p = 4n + 3$, $p \neq a^2 + b^2$.

Necessity: $\forall \pi \in \mathbb{Z}[i]$ such that π is a Gaussian prime. We can factorize

$$N(\pi) = \pi \bar{\pi} = \prod_{i=1}^r p_i^{\alpha_i}$$

. Thus, $\exists p_i$, assuming it to be p_1 , such that $\pi \mid p_1$. Then, $N(\pi) \mid N(p_1) = p_1^2$, which implies $N(\pi) \mid p_1, p_1^2$. If $N(\pi) \mid p_1$, the π is in the first and second case. Else-wise, the $N(\pi) = p_1^2$ and $\pi \mid p_1 \implies \pi \sim p_1$ and p_1 is non-factorized, which is exactly the third case. $\square$

Theorem 3.4. For $\mu = a + bi$, the Set

$$S = \left\{ m + ni \mid 0 \leq m \leq \frac{N(\mu)}{(a, b)} - 1, 0 \leq n \leq (a, b) - 1 \right\}$$

is modulo μ 's complete system of residues in $\mathbb{Z}[i]$.

Proof. Theorem 3.4 is equivalent to

$$\begin{cases} (1) : \forall \alpha, \beta \in S, \alpha \not\equiv \beta \pmod{\mu} \\ (2) : \forall \alpha \in \mathbb{Z}[i], \exists \beta \in S, \alpha \equiv \beta \pmod{\mu} \end{cases}$$

(1): For

$$m + ni \equiv m' + n'i \pmod{\mu},$$

$\mu \mid (m - m') + (n - n')i$. By theorem 2.4, we know that $(a, b) \mid (n - n')$, and thus $n = n'$. Therefore,

$$m - m' \mid \theta = \frac{CN(\mu)}{(a, b)} \implies m = m'.$$

(2): If $\alpha = c + di \in \mathbb{Z}[i]$,

$$d = q(a, b) + r, q \in \mathbb{Z},$$

and

$$c - q(ax_0 - by_0) = \frac{q'N(\mu)}{(a, b)} + r', 0 \leq r' \leq \frac{N(\mu)}{(a, b)} - 1,$$

where $ax_0 - by_0$ is the same as theorem 2.4. Once we combine the two equation:

$$c + di = \frac{q'N(\mu)}{(a, b)} + q(ax_0 - by_0) + q(a, b)i + r' + ri$$

$$c + di = \theta + r' + ri$$

$\square$